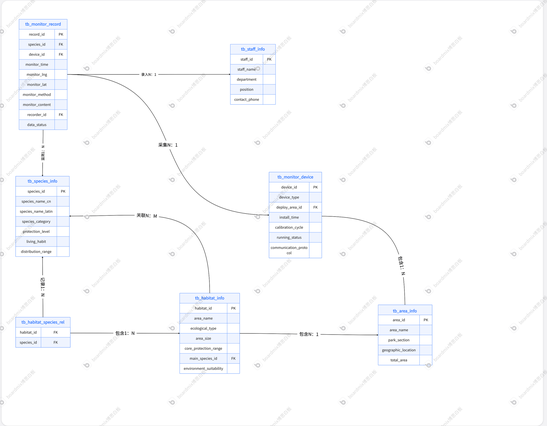
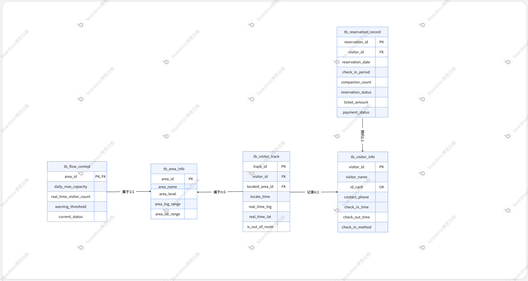
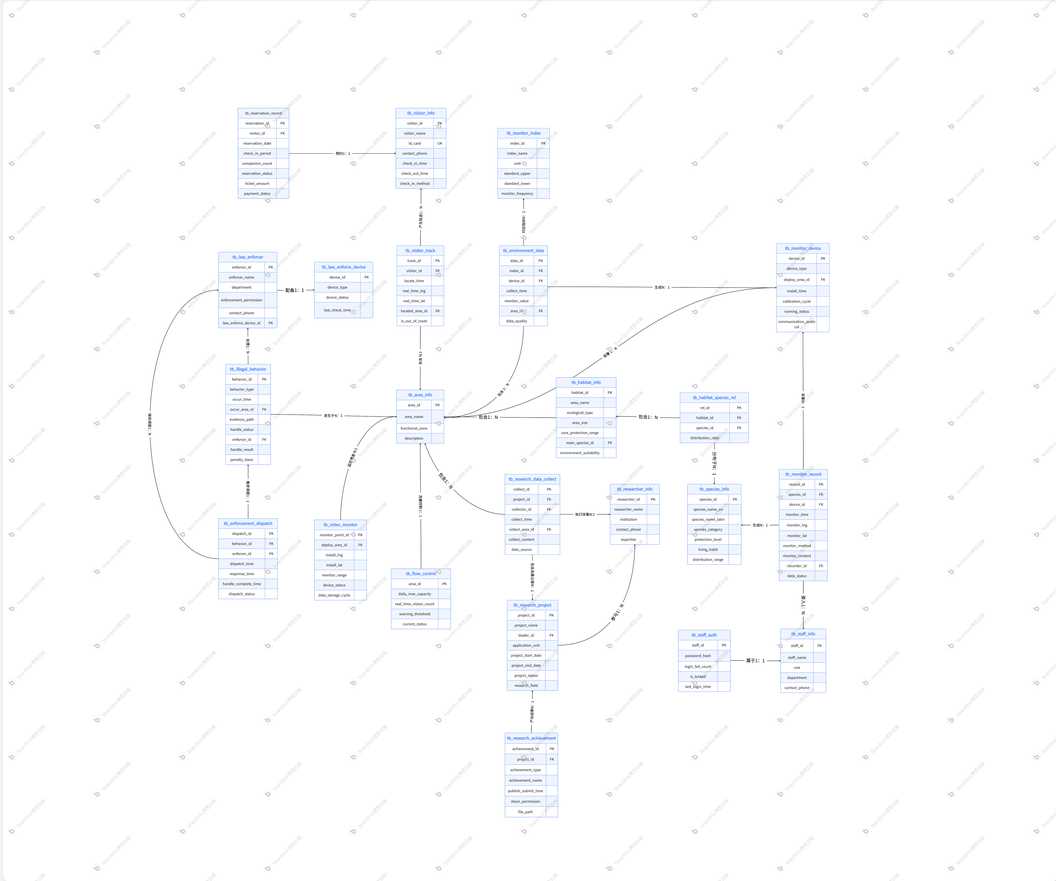
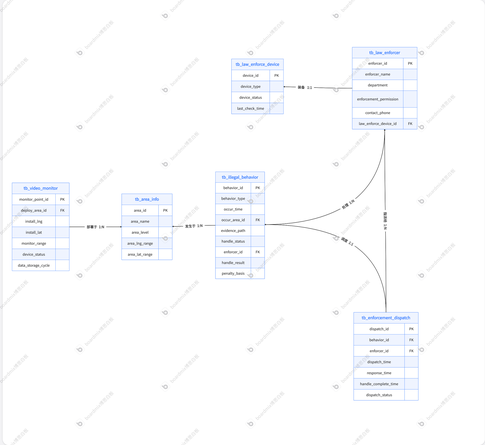

tb_monitor_record
record_id PK
species_id FK
device_id FK
monitor_time
monitor_lng
monitor_lat
monitor_method
monitor_content
recorder_id FK
data_status
录入N: 1
tb_staff_info
staff_id PK
staff_name
department
position
contact_phone
采集N: 1
属于1: N
tb_species_info
species_id PK
species_name_cn
species_name_latin
species_category
protection_level
living_habit
distribution_range
关联M: M
tb_monitor_device
device_id PK
device_type
display_area_id FK
install_time
calibration_cycle
running_status
communication_protocol
位于N: 1
tb_habitat_species_rel
habitat_id FK
species_id FK
包含1: N
tb_habitat_info
habitat_id PK
area_name
ecological_type
area_size
core_protection_range
main_species_id FK
environment_suitability
包含N: 1
tb_area_info
area_id PK
area_name
park_section
geographic_location
total_area


tb_reservation_record
reservation_id PK
visitor_id FK
reservation_date
check_in_period
companion_count
reservation_status
ticket_amount
payment_status
产生N: 1
tb_flow_control
area_id PK, FK
daily_max_capacity
real_time_visitor_count
warning_threshold
current_status
属于1: 1
tb_area_info
area_id PK
area_name
area_level
area_lng_range
area_lat_range
属于N: 1
tb_visitor_track
track_id PK
visitor_id FK
located_area_id FK
locate_time
real_time_lng
real_time_lat
is_out_of_round
记录N: 1
tb_visitor_info
visitor_id PK
visitor_name
id_card UK
contact_phone
check_in_time
check_out_time
check_in_method

tb_area_info
area_id PK
area_name
area_level
area_lng_range
area_lat_range
project_level
部署1: N
tb_monitor_device
device_id PK
device_type
display_area_id FK
install_time
calibration_cycle
running_status
communication_protocol
安装1: N
tb_environment_data
data_id PK
index_id FK
device_id FK
collect_time
monitor_value
area_id FK
data_quality
关联1: N
tb_monitor_index
index_id PK
index_name
unit
standard_upper
standard_lower
monitor_frequency


tb_reservation_record
reservation_id PK
visitor_id FK
reservation_date
check_in_period
companion_count
reservation_status
ticket_amount
payment_status
预约1: 1
tb_visitor_info
visitor_id PK
visitor_name
id_card UK
contact_phone
check_in_time
check_out_time
check_in_method
产生轨迹N: 1
tb_monitor_index
index_id PK
index_name
unit
standard_upper
standard_lower
monitor_frequency
关联指标N: 1
tb_law_enforcer
enforcer_id PK
enforcer_name
department
enforcement_permission
contact_phone
law_enforce_device_id FK
配备1: 1
tb_law_enforce_device
device_id PK
device_type
device_status
last_check_time
tb_visitor_track
track_id PK
visitor_id PK
locate_time
real_time_lng
real_time_lat
located_area_id FK
is_out_of_room
记录N: 1
tb_environment_data
data_id PK
index_id FK
device_id FK
collect_time
monitor_value
area_id FK
data_quality
关联1: N
tb_monitor_device
device_id PK
device_type
display_area_id FK
install_time
calibration_cycle
running_status
communication_protocol
安装N: 1
部署1: N
tb_illegal_behavior
behavior_id PK
behavior_type
occur_time
occur_area_id FK
evidence_path
handle_status
enforcer_id FK
handle_result
penalty_basis
执行N: 1
tb_area_info
area_id PK
area_name
functional_name
description
包含N: 1
tb_habitat_info
habitat_id PK
area_name
ecological_type
area_size
core_protection_range
main_species_id FK
environment_suitability
包含1: N
tb_habitat_species_rel
rel_id PK
habitat_id FK
species_id FK
distribution_range
属于1: N
属于N: 1
关联N: 1
关联N: 1
关联N: 1
tb_enforcement_dispatch
dispatch_id PK
behavior_id FK
enforcer_id FK
dispatch_time
response_time
handle_complete_time
dispatch_status
tb_video_monitor
monitor_point_id PK
display_area_id FK
install_lng
install_lat
monitor_range
device_status
data_storage_cycle
关联监控N: 1
tb_research_data_collect
collect_id PK
project_id FK
collector_id FK
collect_time
collect_area_id FK
collect_content
data_source
执行采集N: 1
tb_researcher_info
researcher_id PK
researcher_name
institution
contact_phone
expertise
tb_species_info
species_id PK
species_name_cn
species_name_latin
species_category
protection_level
living_habit
distribution_range
关联N: 1
tb_monitor_record
record_id PK
species_id FK
device_id FK
monitor_time
monitor_lng
monitor_lat
monitor_method
monitor_content
recorder_id FK
data_status
录入N: 1
tb_flow_control
area_id PK
daily_max_capacity
real_time_visitor_count
warning_threshold
current_status
tb_research_project
project_id PK
project_name
leader_id FK
application_unit
project_start_date
project_end_date
project_status
research_field
参与N: 1
参与N: 1
tb_research_achievement
achievement_id PK
project_id FK
achievement_type
achievement_name
publish_submit_time
share_permission
file_path
产出N: 1
tb_staff_auth
staff_id PK
password_hash
login_fail_count
is_locked
last_login_time
属于1: 1
tb_staff_info
staff_id PK
staff_name
role
department
contact_phone
录入N: 1


tb_law_enforce_device
device_id PK
device_type
device_status
last_check_time
属于 1:1
tb_law_enforcer
enforcer_id PK
enforcer_name
department
enforcement_permission
contact_phone
law_enforce_device_id FK
tb_video_monitor
monitor_point_id PK
display_area_id FK
install_lng
install_lat
monitor_range
device_status
data_storage_cycle
部署于 1:N
tb_area_info
area_id PK
area_name
area_level
area_lng_range
area_lat_range
发生于 1:N
tb_illegal_behavior
behavior_id PK
behavior_type
occur_time
occur_area_id FK
evidence_path
handle_status
enforcer_id FK
handle_result
penalty_basis
处理 1:N
监督 1:N
tb_enforcement_dispatch
dispatch_id PK
behavior_id FK
enforcer_id FK
dispatch_time
response_time
handle_complete_time
dispatch_status

tb_researcher_info
researcher_id PK
researcher_name
affiliated_unit
research_field
contact_info
主持项目
参与项目
tb_research_achievement
achievement_id PK
project_id FK
achievement_type
achievement_name
publish_submit_time
share_permission
file_path
产出成果
tb_research_project
project_id PK
project_name
leader_id FK
application_unit
project_start_date
project_end_date
project_status
research_field
包含N:1
tb_research_data_collect
collect_id PK
project_id FK
collector_id FK
collect_time
collect_area_id FK
collect_content
data_source
采集N:1
tb_area_info
area_id PK
area_name
area_level
area_lng_range
area_lat_range